DEPARTMENT OF COMPUTER SCIENCE
& INFORMATION TECHNOLOGY

CT-469 · REINFORCEMENT LEARNING COURSE PROJECT BRIEF

Spring 2026 | Final Year – AI Specialisation

Course Code	CT-469
Project Weight	10 Marks (10% of Total Grade)
Team Size	3 Members per Group
Evaluation Week	Week 14/15
AI Tools Policy	Permitted with mandatory disclosure

"Build real systems that solve real problems using Reinforcement Learning"

1. PROJECT OVERVIEW & PHILOSOPHY

This course project is your opportunity to move beyond theory and build a working, deployable prototype that uses Reinforcement Learning to address a genuine real-world problem. This is a collaborative that mirrors industry practice.

The project is intentionally open-ended. You are asked to design, build, and demonstrate a system. AI tools are explicitly permitted because modern software engineers use them routinely. What matters is your ability to understand the RL mechanics you have implemented, justify your design decisions, and demonstrate that your system actually works.

1.1 What Makes a Good Project

Dimension	Strong Project	Weak Project
Problem Relevance	Solves a specific, named real world problem	Demonstrates an algorithm on a toy environment only
RL Integration	RL is the core decision-making engine	RL is a minor add-on to an existing system
Full-Stack Depth	Complete prototype: UI + backend + RL + data	Only a Jupyter notebook or CLI script
Comparison	RL compared with at least one baseline method	Only RL, no comparison or evaluation

Demonstration	Live working demo, training curves, analytics	Slides only, no runnable system

1.2 AI Tool Policy

This project explicitly permits the use of AI coding tools (GitHub Copilot, ChatGPT, Claude, etc.). The following rules apply:

✅ Permitted	❌ Not Permitted
Generate boilerplate UI / API code	Copy an entire RL implementation without understanding it
Debug errors and fix syntax	Submit AI-generated reports as your own writing
Generate documentation and README	Use AI to answer viva questions during evaluation
Use AI to explain RL concepts for study	Misrepresent AI contributions in your disclosure

Every group must submit an AI Usage Disclosure (template provided in Section 5). Groups that use AI transparently will not be penalised. Groups that misrepresent their AI usage will lose marks.

2. DELIVERABLES & SUBMISSION REQUIREMENTS

All groups must submit the following four deliverables by Week 14. Submissions are through the course google classroom unless otherwise instructed.

2.1 Deliverable 1 — Working Prototype (4 marks)

A functional, runnable software prototype that demonstrates the RL system solving the target problem. The prototype must be self-contained and deployable from your repository with minimal setup.

Requirement	Details
Source Code Repository	GitHub / GitLab repository with commit history. Must show iterative development — not a single bulk commit.
README.md	Step-by-step setup guide, dependency list, how to run training, how to run the web app, how to reproduce results.
Web Interface	A user-facing web application (not just API endpoints or notebooks). Must be accessible via browser.
RL Training Script	Standalone Python script or notebook that trains the RL agent from scratch. Must log training metrics.
Saved Model	Pre-trained model weights included in the repository so evaluators can run the demo without waiting for training.
Baseline Implementation	At least one non-RL baseline method implemented and comparable under the same evaluation conditions.

Environment / Simulation	A custom or adapted environment that models the real-world problem. Must be reproducible with a fixed random seed.

2.2 Deliverable 2 — Project Report (2 marks)

A technical report (5–10 pages) documenting your system design, implementation decisions, experimental results, and critical analysis.

2.3 Deliverable 3 — Presentation & Live Demo (3 marks)

A 15-minute group presentation with a live demonstration of the working prototype, followed by a 10- minute Q&A with the instructor.

Important: All team members must present and must be able to answer technical questions independently. A member who cannot answer questions about the RL component will have their individual mark reduced.

2.4 Deliverable 4 — Individual Reflection (1 mark)

Each team member contributions should be clearly at the start of project report.

1. What was your specific individual contribution to the project? Be precise — list files, functions, or components you built.
2. What was the most difficult technical challenge you personally faced? How did you resolve it?
3. How did you use AI tools in this project? Be specific about what you prompted, what you accepted, and what you changed.
4. What is one thing you now understand about RL that you did not understand at the start of the project?

3. AI USAGE DISCLOSURE FORM

Complete and submit this form as an appendix to your project report. Honest disclosure is rewarded — it will not reduce your mark.

AI Usage Disclosure Form — CT-469 Project
Group Name / Number: ________________________ Project Title: ________________________ Member 1: ________________________ Member 2: ________________________

Project Component	AI Tool Used	How It Was Used	% AI-generated
Frontend / UI			
Backend / API			
RL Environment			

RL Agent / Training			
Report Writing			
Debugging			
Documentation			

Signature: Member 1 ________________________ Member 2 ________________________

4. EVALUATION CRITERIA FOR AI-ASSISTED PROJECTS

Since AI tools are permitted, evaluation focuses on depth of understanding rather than volume of code written. The following criteria distinguish high-quality from low-quality AI-assisted work:

Dimension	High Quality (AI used wisely)	Low Quality (AI overused)
RL Design	Reward function reflects deep problem understanding. MDP choices are justified by domain knowledge.	Generic reward (+1/-1). MDP copied from a tutorial with no problem-specific adaptation.
Code Quality	Readable, modular, well commented. Commits show iterative thinking. Can explain every function.	Monolithic, hard to follow. Single commit. Cannot explain own code in Q&A.
Results Analysis	Honest analysis: explains failures, discusses why RL outperforms / underperforms baseline.	Only shows success cases. No discussion of when system fails.
System Design	Architecture is tailored to the problem. Trade-offs are explicitly justified.	Generic template architecture copied with no problem-specific customisation.
Q&A Performance	Can derive equations, explain decisions, modify code live, predict effects of changes.	Cannot explain code. Defers to 'the AI wrote it'. Cannot handle follow-up questions.

Build something you are proud to demo. Good luck.